

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.6_1)

Metric

$$ds^2 = \frac{3}{2\Lambda y^2} du \otimes dv + \frac{3}{2\Lambda y^2} dv \otimes du + \frac{3}{\Lambda y^4} dv \otimes dv - \frac{3}{\Lambda y^2} dy \otimes dy - \frac{3}{\Lambda y^2} dz \otimes dz$$

Coordinates

$$(x^\mu) = (u, v, y, z)$$

Null Tetrad

Covariant components

$$l = (1) \, dv$$

$$n = \left(-\frac{3}{2\Lambda y^2}\right) du + \left(-\frac{3}{2\Lambda y^4}\right) dv$$

$$m = \left(\frac{1}{2}\sqrt{3}\sqrt{-\frac{1}{\Lambda y^2}}\right) dy + \left(\frac{1}{2}i\sqrt{3}\sqrt{-\frac{1}{\Lambda y^2}}\right) dz$$

$$\bar{m} = \left(\sqrt{3}\sqrt{-\frac{1}{\Lambda y^2}}\right) dy + \left(-i\sqrt{3}\sqrt{-\frac{1}{\Lambda y^2}}\right) dz$$

Contravariant components

$$l = \left(\frac{2}{3}\Lambda y^2\right) du$$

$$n = \left(\frac{1}{y^2} \right) du + (-1) dv$$

$$m = \left(-\frac{\sqrt{3}\Lambda|y|}{6\sqrt{-\Lambda}} \right) dy + \left(-\frac{i\sqrt{3}\Lambda|y|}{6\sqrt{-\Lambda}} \right) dz$$

$$\bar{m} = \left(-\frac{\sqrt{3}\Lambda|y|}{3\sqrt{-\Lambda}} \right) dy + \left(\frac{i\sqrt{3}\Lambda|y|}{3\sqrt{-\Lambda}} \right) dz$$

Directional Derivatives

$$D = \frac{2}{3} \Lambda y^2 \partial_u$$

$$\Delta = \frac{1}{y^2} \partial_u - \partial_v$$

$$\delta = -\frac{\sqrt{3}\Lambda|y|}{6\sqrt{-\Lambda}} \partial_y - \frac{i\sqrt{3}\Lambda|y|}{6\sqrt{-\Lambda}} \partial_z$$

$$\bar{\delta} = -\frac{\sqrt{3}\Lambda|y|}{3\sqrt{-\Lambda}} \partial_y + \frac{i\sqrt{3}\Lambda|y|}{3\sqrt{-\Lambda}} \partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = -\frac{\sqrt{3}}{\sqrt{-\Lambda}y^3|y|}$$

$$\alpha = \frac{\sqrt{3}\sqrt{-\Lambda}y}{3|y|}$$

$$\beta = 0$$

$$\tau = -\frac{\sqrt{3}\Lambda|y|}{6\sqrt{-\Lambda}y}$$

$$\pi = \frac{\sqrt{3}\Lambda|y|}{3\sqrt{-\Lambda}y}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{5}{2y^4}$$

$$\Lambda = \frac{1}{6}\Lambda$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{3}{y^4}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "N"